Matias Andia

mati@princeton.edu | github.com/matias-ma

Education

Princeton University Princeton, New Jersey
A.B. in Mathematics Expected May 2027

Selected coursework: Numerical analysis, complex analysis, measure theory, and graduate courses in functional analysis, quantum field theory, and computational cryo-EM.

Research Interests

Computational and applied mathematics, numerical linear algebra, tensor methods, high-dimensional computation, and high-energy physics.

Research Experience

Tensor decompositions and density estimation May 2025–Present
Princeton University
Advisor: Prof. Marc Aurèle Gilles

- Adapted and implemented tensor-based density estimation methods to mitigate the curse of dimensionality in reconstructing conformational heterogeneity in cryo-electron microscopy.
- Developed algorithms for tensor decompositions of multivariate functions with rigorous convergence rates.

High-energy astrophysics August 2023–Present
Princeton University, Lisanti Group
Advisor: Prof. Carlos Blanco (Penn State)

- Demonstrated that current predictions overestimate the contribution of unresolved TeV blazars to the isotropic gamma-ray background; manuscript in preparation.

Neutrino physics June 2022–August 2023
Harvard University, Argüelles-Delgado Group, IceCube Neutrino Observatory
Advisor: Julia Book Motzkin

- Assisted in a search for heavy neutral leptons using a water Cherenkov detector.

Preprints

[1] Matias Andia. *A Hadamard matrix with growth factor greater than its order.* Zenodo preprint, 2026. DOI: 10.5281/zenodo.23871097.

[2] Matias Andia. *A completely pivoted Hadamard matrix with a non-final pivot equal to n.* Zenodo preprint, 2026. DOI: 10.5281/zenodo.23894280.

Poster Presentations

A functional tensor train by recursive Gaussian elimination 2026
11th International Conference on Curves and Surfaces.

Tensor methods for density estimation 2025
Computational Cryo-EM Workshop, Flatiron Institute.

Tensor density estimation in cryo-EM 2025
SIAM New York–New Jersey–Pennsylvania Section Conference.

Teaching Experience

Undergraduate Course Assistant September–December 2025
Princeton University
MAT 321: Numerical Analysis and Numerical Linear Algebra
Professor: Marc Aurèle Gilles

- Supported students with problem sets and course material and held office hours.
- Reviewed draft exams and offered feedback to the professor.

Tutor September 2024–Present
McGraw Center for Teaching and Learning, Princeton University

- Tutor students in an economics-based statistics course.

Research Funding

Undergraduate Math Research Funding 2025, 2026
Department of Mathematics, Princeton University.

Undergraduate Fund for Academic Conferences (UFAC) 2025, 2026
Office of Undergraduate Research, Princeton University.

PACM research funding 2025
Program in Applied and Computational Mathematics, Princeton University.

OURSIP research funding 2024
Office of Undergraduate Research, Princeton University.

Computing Skills

Python (advanced); Julia and Mathematica (proficient).
Bash and SLURM (proficient).